Sequences, Series, and the Binomial Theorem

Evaluating binomial coefficients, expanding a binomial power in full, extracting one specific term, and reading the structure of a row of coefficients

Directions. The problems below rotate between four tools, so decide which one the question needs before you start writing.

$$(p+q)^n = \sum_{r=0}^{n} \binom{n}{r} p^{\,n-r} q^{\,r}, \qquad \binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

- A single term needs the coefficient *and* both powers — the binomial coefficient by itself is almost never the answer.
- Simplify factorials by cancelling before multiplying.

Reference. The first rows of Pascal's triangle, shown for their pattern only. No problem's numbers are taken from this display; use the values each problem gives you.

$$\begin{array}{ccccccccc} & & & & 1 & & & & \\ & & & 1 & & 1 & & & \\ & & 1 & & 2 & & 1 & & \\ & 1 & & 3 & & 3 & & 1 & \\ 1 & & 4 & & 6 & & 4 & & 1 \end{array}$$

1. Warm-up. Evaluate $\binom{7}{3}$ by cancelling factorials before you multiply. Show the cancellation.

Answer: ________________

2. Warm-up. Evaluate $\binom{9}{2}$. A classmate answers 72. Say what 72 actually counts, and give the correct value of $\binom{9}{2}$.

Answer: __________________

3. Full expansion. Expand $(x+3)^3$ completely, using the binomial theorem rather than repeated multiplication. Write the four terms in descending powers of x.

Answer: __________________

4. One term only. In the expansion of $(x+4)^5$, find the coefficient of the term containing x^2. Do not expand the whole binomial — name the value of r you need first.

Answer: ________________

5. Row structure. Find the sum of all the entries in row 6 of Pascal's triangle (the row whose second entry is 6). Then explain, using the binomial theorem, why substituting 1 for both letters in $(p+q)^n$ gives that sum.

Answer: ________________

6. Full expansion. Expand $(2x-1)^4$ completely. Watch both the power on the 2 and the alternating sign.

Answer: __________________

7. One term only. In the expansion of $(3+y)^6$, find the coefficient of y^4. State which factor of the term the exponent on 3 comes from.

Answer: __________________

8. Solve for n. A tournament pairs every team with every other team exactly once, so the number of games is $\binom{n}{2}$, which equals $\dfrac{n(n-1)}{2}$. Solve $\dfrac{n(n-1)}{2} = 45$ for n. Give both solutions of the equation, then say which one can be a number of teams and why.

Answer: __________________

9. Row structure. Find the sum of the coefficients of $(1+2x)^5$ without expanding it. State the substitution that produces the sum of the coefficients of any polynomial.

Answer: __________________

10. Full expansion, two letters. Expand $(a+2b)^3$ completely, in descending powers of a. Keep the power of 2 attached to the correct factor in every term.

Answer: ________________

11. One term only. In the expansion of $(2x+y)^5$, find the coefficient of the term x^3y^2. Say why the coefficient is not simply the entry from Pascal's triangle.

Answer: ________________

12. Synthesis. Consider $(3x-2)^4$.

Find the sum of its coefficients using the same substitution as before, and then explain what that sum tells you about the value of the polynomial at a particular input. Finish by saying why the answer is so much smaller than the individual coefficients of the expansion.

Answer: ________________